

Multi-Modal Accessibility to Key Services in the Maldives

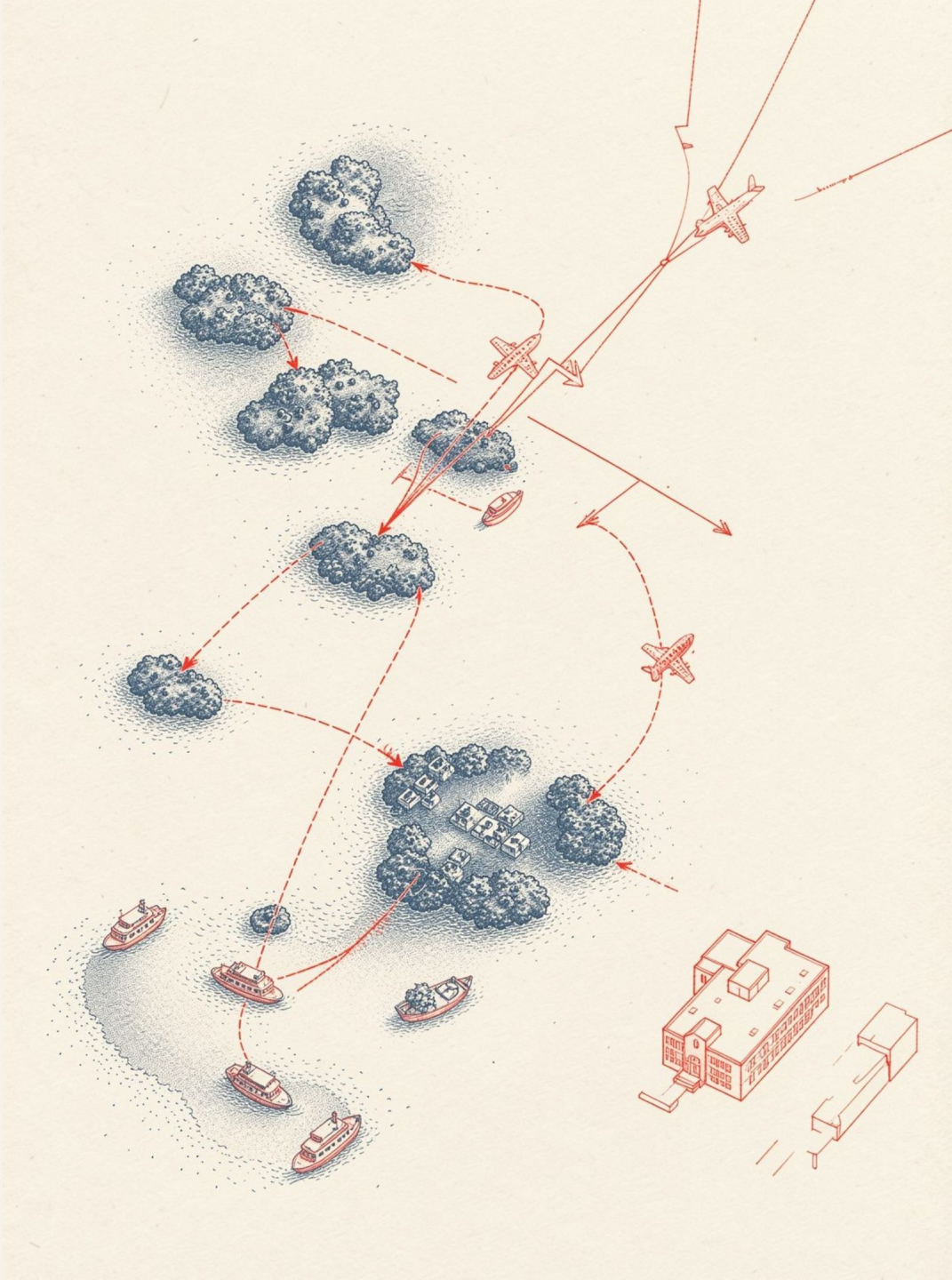
Datathon 2026 · [A Team]

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Problem Statement

Access to key services such as universities and primary healthcare facilities remains uneven across Maldives due to geographical fragmentation. Many islands lack convenient and affordable transport links, limiting equitable access to essential services.

- Access to key services uneven across the Maldives.
- Fragmented geography; islands lack affordable, convenient transport.
- Essential services out of reach for many island populations.



Key Stats

01 **Student access gap:** 48% of atoll grade 9–10 students live on islands without grades 11–12 (MoE/MBS School Statistics, 2023)

02 **Education-driven migration:** 43–53% of migration to Malé from key atolls is education-related (MBS Census & Migration Data, 2014–2022)

03 **Land-only access:** ~47% of people can reach a school; coverage flat (46–48%) from 10–120 min (HeiGIT/OpenAccessLens, HDX)

04 **Specialist coverage:** Many atolls have no tertiary-level specialist services, requiring inter-island/abroad referral (MoH HEMS / Aasandha reports)

Implementation



Data Exploration



Geo-spatial preparation



POI mapping & Network construction



Accessibility analysis



Data visualization

Demo

Impact & Limitations

Impact:

Help planners, education/health ministries, atoll councils — prioritize routes, place services, target subsidies.

Limitation:

Data completeness (schedules, facility coordinates, fares).

References & Acknowledgements

Data: MBS (Census 2022, Datathon datasets), RTL/MTCC, MACL, FollowMe, Avas Ride, Human Development Exchange (HDX)	**Research:** HeiGIT/OpenAccessLens (HDX), OpenStreetMap
Platform: Python, Django, htmx, Leaflet + custom GeoJSON, hand-written CSS, SQLite	**Tools:** pandas, NumPy, openpyxl
Presentation: Gamma	**AI Usage:** ChatGPT & OpenCode (for planning), Claude (for coding)

Thank you

Any Questions